

Zaira Shahid

AI Automation Engineer & Full-Stack Engineer

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PROFESSIONAL PROFILE

AI Automation Engineer and Software Engineering graduate with hands-on experience building RAG-based LLM chatbots, voice AI agents, and computer vision systems for real-world clients. Delivered a production-deployed Credit Card Fraud Detection System, client-facing RAG chatbots that reduced manual query handling time, and a real-time Driver Drowsiness Detection System. Experienced across the full-stack (MERN) and in end-to-end ML pipeline development, REST API deployment, and cloud production environments. Seeking an AI Automation Engineer or Full-Stack Engineer role to design and ship intelligent, production-grade systems.

TECHNICAL SKILLS

AI & LLMs: RAG, LLMs, Prompt Engineering, Conversational AI, LangChain, LangGraph, Scikit-learn, Random Forest, Naive Bayes, Computer Vision (OpenCV, YOLOv8, MediaPipe).

Development: Python, FastAPI, React, Next.js, Node.js, Express.js, MongoDB, MySQL, Supabase, n8n (Workflow Automation).

Tools & DevOps: Docker, Railway, Vercel, Firebase, Git, GitHub, VS Code, Kaggle, Jupyter Notebook.

PROFESSIONAL EXPERIENCE

AI Automation Developer

Sovanza

Faisalabad, Pakistan

2024 – 2026

- Developed RAG-based LLM chatbots for 5+ client businesses using Retrieval-Augmented Generation, integrating custom knowledge bases to cut manual customer query handling time by an estimated 30%.
- Engineered Voice AI automation agents using Vapi for client-facing lead qualification and support, handling conversational flows for multiple production deployments.
- Managed Supabase backend including database schema design, REST API integration, and real-time data handling across 3+ client projects.
- Applied prompt engineering techniques and LLM integration to build end-to-end AI automation workflows, cutting manual workflow steps by an estimated 40% for clients.

MERN Stack Engineer

Excellium

2023 – 2024

- Delivered 4+ full-stack web applications using MongoDB, Express.js, React, and Node.js (MERN stack) as part of a 3–5 person engineering team.
- Designed and integrated 10+ RESTful API endpoints to connect front-end interfaces with backend services and databases, improving data-handling reliability across modules.
- Collaborated with a 3–5 person cross-functional team to design responsive, user-friendly web interfaces, improving page load speed by an estimated 20%.

KEY PROJECTS

AI Social Media Automation System | Next.js, TypeScript, Firebase, n8n, Groq LLM, Cloudinary, Slack [GitHub](#)

Built an end-to-end AI automation suite covering content generation, scheduling/publishing, engagement automation, and analytics across Facebook, Instagram, and LinkedIn; designed the workflow pipeline (research → shortlist → human approval → generate → schedule) using n8n and LLM-based content generation, architected entirely on free-tier infrastructure.

Menstrual & Women's Safety Support System (Final Year Project) | Bilingual (English/Urdu) safety app, AI chatbot, Supabase [GitHub](#)

Built a bilingual (English/Urdu) mother and women's safety application featuring an AI chatbot with text and voice interaction and a clinic-finder map; integrated NGOs and pharmacies into the platform to connect users with real-world support and resources.

Gold Trading Bot | Next.js, Firebase, n8n, Vercel, Telegram API [GitHub](#)

Automated a gold paper-trading simulation with a real-time Next.js dashboard and Firebase backend, tracking trades across multiple sessions; integrated n8n workflows with Telegram to deliver automated trade signal generation and real-time alerts.

Driver Drowsiness Detection System | Python, OpenCV, Tkinter, PyInstaller [GitHub](#)

Engineered a real-time computer vision safety system achieving 95%+ drowsiness-detection accuracy via webcam-based eye tracking; implemented a blink-rate analysis algorithm triggering an audio alarm, and packaged it as a standalone executable for non-technical end users.

Credit Card Fraud Detection System | *Python, Scikit-learn, FastAPI, Docker, Railway, Vercel, Twilio*

Developed an end-to-end ML pipeline using a Random Forest Classifier on 50,800 transactions, achieving 100% accuracy using SMOTE oversampling; implemented multi-layer fraud prevention with real-time risk scoring and OTP-based MFA via Twilio Verify API, deployed via FastAPI/Docker on Railway with a real-time analytics dashboard on Vercel.

NovaBot | *Llama 3.3/4 (Groq API), PDF.js, Web Speech API, JavaScript* [GitHub](#)

Built a fully client-side multimodal AI assistant with vision, voice, and document understanding — zero backend, deployed live on Vercel; integrated Llama 3.3 70B for text and Llama 4 Scout for vision via the Groq API, with PDF.js for document reading and the Web Speech API for voice input.

EDUCATION

BS Software Engineering — NUML University, Faisalabad, Pakistan | CGPA: 3.7